

Dissecting Model Visuals Transcript

Hi everyone, welcome to BITS Pilani digital programme. Today, we will be discussing how to dissect model visuals. In this process, you are actually dissecting the visualisations with a singular purpose to improve the effectiveness of communications.

In all this you have to ask what is the intended purpose of the visualisation, it is to communicate the information in most effective format and trying to make the audience understand your point of view. So, as to enable them to take better decisions. However, we will also see the use of visual hierarchies, clearing the axis and adding some more visual elements to the existing visual in developing the visualisation or improving its form and factor.

These steps form the dissecting phase of model visuals, these steps form part of dissecting the model visuals. This will help you to improve the content within your visualisation. We will now do this by looking at a series of examples and develop the visual, so that it communicates the information in a more effective format.

The first example that we will be looking today is a line graph. In this case, the graph simply explains the comparison between the campaign fundraising between last year and the current year. So, it is clearly showing the growth between days since the campaign has launched about and it is tracking the monthly progress.

If you look at the graph on the slide, you are plotting two things, you are simply plotting the days since the campaign has begin and how much the funds are raised in this year as compared to the last year. If you draw a simple line graph, it would be simply two lines. However, this would not present the adequate information.

Now look at the graphic that is there on the slide. The graph has clearly portrayed the last year funding line in a light blue colour as compared to the current year funding line in a thick blue line. This shows the current year is more important and more relevant information to the audience, whoever it might be, because the purpose is to show the progress, compare the progress between the current year and the last year.

And it shows clearly what is the goal, 50,000 dollars is the goal. And last year, they have surpassed it almost at the end of the campaign period and currently they are at a higher pace, relatively higher pace as compared to last year and they have collected already 33,967 dollars. So, in this way, the graphic is able to capture 3, 4 things in one shot and also to show it effectively.

It is not simply using the same colour for the both the lines, it is using a different colour by bringing more attention to the current year. And also it has added a small data point at the end of the first year, at the end of the current year showing the amount that is collected till now compared to the amount that is collected in the overall campaign period in the last year. And also clearly the goal post is given as a horizontal bar in the graph showing that this is the goal and the gap to the goal is clearly visible.

Also given the relative slope of the current year slide, you can say that pace of campaign fundraising is at a higher rate in the current year as compared to the last year for the same amount of time. Last year at this same period, you are roughly collected less than 20,000 dollars, now we have collected around 33,967 dollars. All this information can be presented in a blunt form or a bland form, but by using the visual aspects like highlighting the current year line with larger thickness as compared to the previous year line which is lighter, you are bringing more perspective to the visual and helping the audience to perceive information in a more efficient manner.

We will now look at the same line graph and for a different purpose. Here the purpose is to forecast the performance of the company or forecast the growth rate that is expected to happen over the next period and see the sales revenue over a period of time. In this graph, you have simply taken the annual numbers, the actual numbers, yearly sales numbers, then you have forecasted using some technique and you have brought your numbers.

Simply it can be a linear forecast, assume that the growth rate is 10% over the year and then you have simply increased the annual revenue and this is the chart. So this chart tries to show the actual numbers and the forecasted numbers. The challenge here is to make audience understand this is a forecast chart, there is a difference between actual and forecast.

The designer could have chosen to simply present everything in a continuous line, but that does not effectively convey the information. So, here the one interesting aspect is the footnote provides the number of projects, so that one can relatively see the actual importance also. Because a stacked graph converts everything into 100%, the quantities are hidden.

So, one might have to compare the quantities, you can add this kind of details because these are not required for everyone, you need not bring them on to the graphic, but you can put them in the footnote, whoever requires that information can refer to the footnote without disturbing the main message that the graphic is trying to convey. This is the process in which you try to dissect the model and understand why certain things are done in a particular form. So, again to reemphasize whenever you are looking at a graphic and there it is well presented, there is a reason why such choices have been made.

Equally, if somebody uses the same colour for missed, meet and exceed and just present the numbers, the effect would have not been this much as compared to using a single attention seeking colour. Also in this particular graphic, the company there is a particular text saying that in quarter 23 about one third of the projects have missed the targets. This is the real concern that the analyst wanted to highlight and by bringing in the text in a selective place and also using the colour which is equivalent to the missed target, the emphasis is directly brought to the point that 42 percent of the projects have missed the target in the current quarter.

This way you can increase the effectiveness of your visualisation by using colour and contrast to your advantage. Now, let us take the example of a same ah stacked bar, but in here there is negative element also which means there is negative growth rate or quantities can go negative. So, we can see overall growth, reduction and increase in the same graphic.

Here we are trying to see expected director population over a period of time. Again you have ah different years ah this is forward planning, you are looking ahead in time. You are today presently in 2015, you are looking 5 years ahead and try to do a human resource plan.

So, again you know what is the unmet gap and you can see what are the reasons, what is the driver for this unmet gap. So, you can see attrition, promotion to directors, directors from acquisitions and the unmet gap as per your business needs. This is the content in the graphic.

Now, let us understand the visual elements here. For example, the colour coded see blue for attrition, green for additions and black for unmet need clearly brings out what is the gap that has to be met. For example, if you see the attrition being in blue and the unmet gap being in black clearly brings out the gap that has to be met.

There is a ordering that has been done and attrition means the people are living leaving the company which means negative flow that is below the x axis. And the gap values are labelled for immediate attention for the persons who are looking at the graphic. Excuse now, coming to the other aspects, coming to other model aspects of this graphic.

For example, there is a logical ordering showing that attrition is at the bottom. Because if there is no attrition, there will not be reduction in numbers. If there is higher attrition, the number would fall and go to the negative territory, right.

You cannot have 0 or negative hires or promoters. It can be 0, but it cannot be negative and the gap can also be on the top. So, there is a logical ordering in the sequence, so that the graph is very intuitively presented.

Both positive and negative directions are there on the graph which is reinforcing the main message that you have an unmet gap of directors that you need to hire or plan out to meet your business expectations in the coming 5 years. Imagine this is being presented to the buff directors, they are now aware of the task that there has to be manpower planning to meet the gap. The graphic is very intuitive in driving 3 points together that there is attrition, the new hiring and promotion is not sufficient and once taken together, there is certain amount of gap that needs to be filled.

This information is clearly conveyed in a single graphic by using the negative axis and colour coding effectively. Now, let us talk about another interesting case of horizontal stacked bars. Here in this case ah they are not scaled to 100 percent, they are simple bars, but presented across category.

This is the data on top 15 developmental priorities according to a survey conducted on roughly about 4000 people and this horizontal layout is readability. Now, had this been a vertical stacked graph, the readability would have been less. So, this what drives the choice of horizontal or vertical bar, you could present the same information with vertical bar if the focus is more on the quantity, but if you are focussing on the top priorities on the causes or the categories that you want to look which are textual is easy to read them in a horizontal bar.

So, you have taken a horizontal bar, the graph is presented horizontally, then you can see the first 3 are the top most priorities soliciting about 45 percent, 37 percent, 32 percent. You are highlighting those numbers in different colours which are then used for the bars also. There is consistency of colour between what is highlighted as development priority and what is used for the colouring of the bar.

Now, you are using ah and one interesting part here is you have omitted the x axis completely because you have used the data labels to indicate the percentages. There is no need to present the entire 100 percent of the x axis because you have maximum 45 percent and you have represented these numbers on the tables itself. So, there is no need for x axis.

You can remove elements that are not necessary in the graphic because you are giving additional or substitute information for the person to read the numbers. So, in this way you can dissect the model visual for the specific purposes and try to impose elements or bring in new elements to enhance the effectiveness of the visual . Finally, there is a footnote which tries to explain the context in which the survey is conducted for people who are interested this information is also available so that it is giving more accessibility and acceptance for the audience.

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